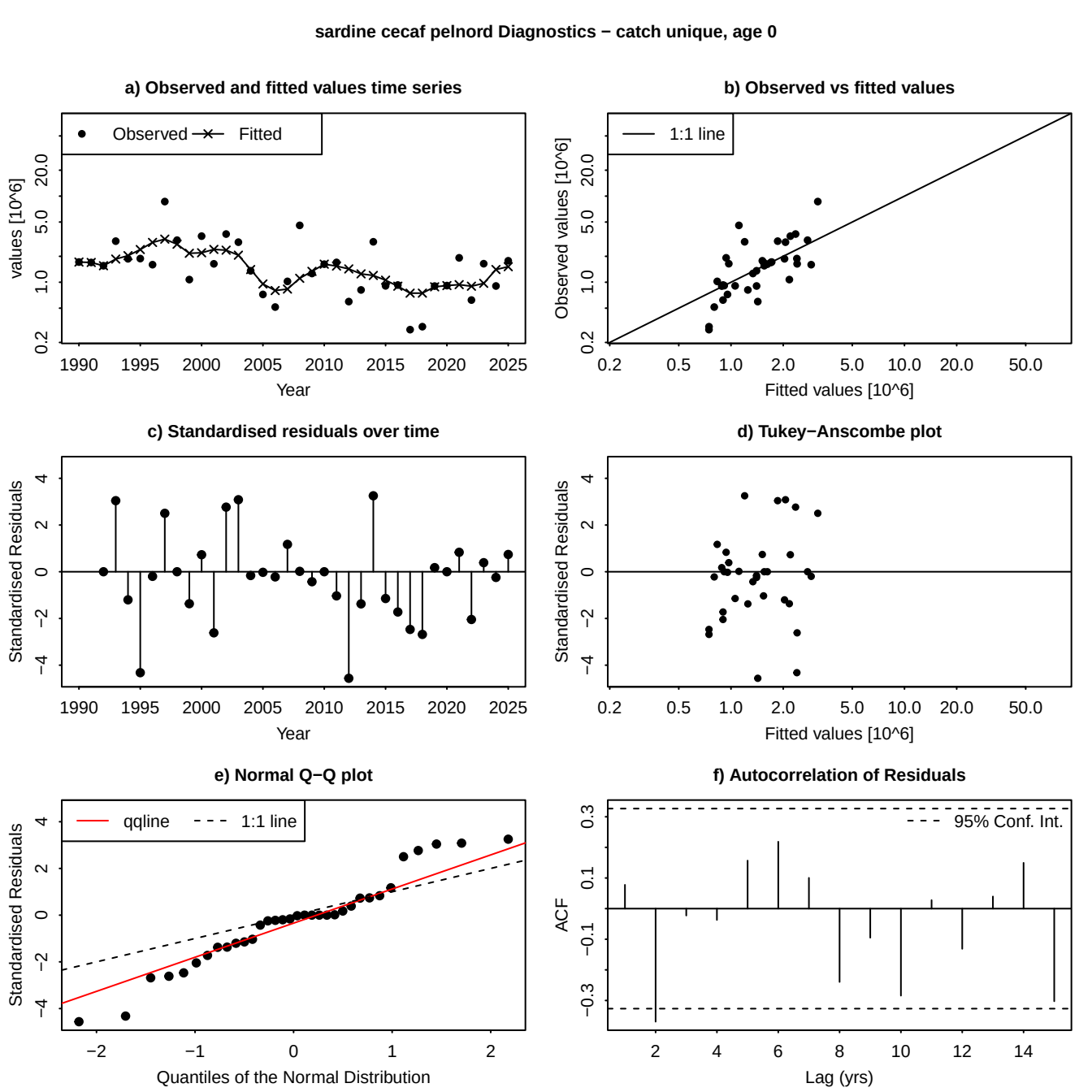
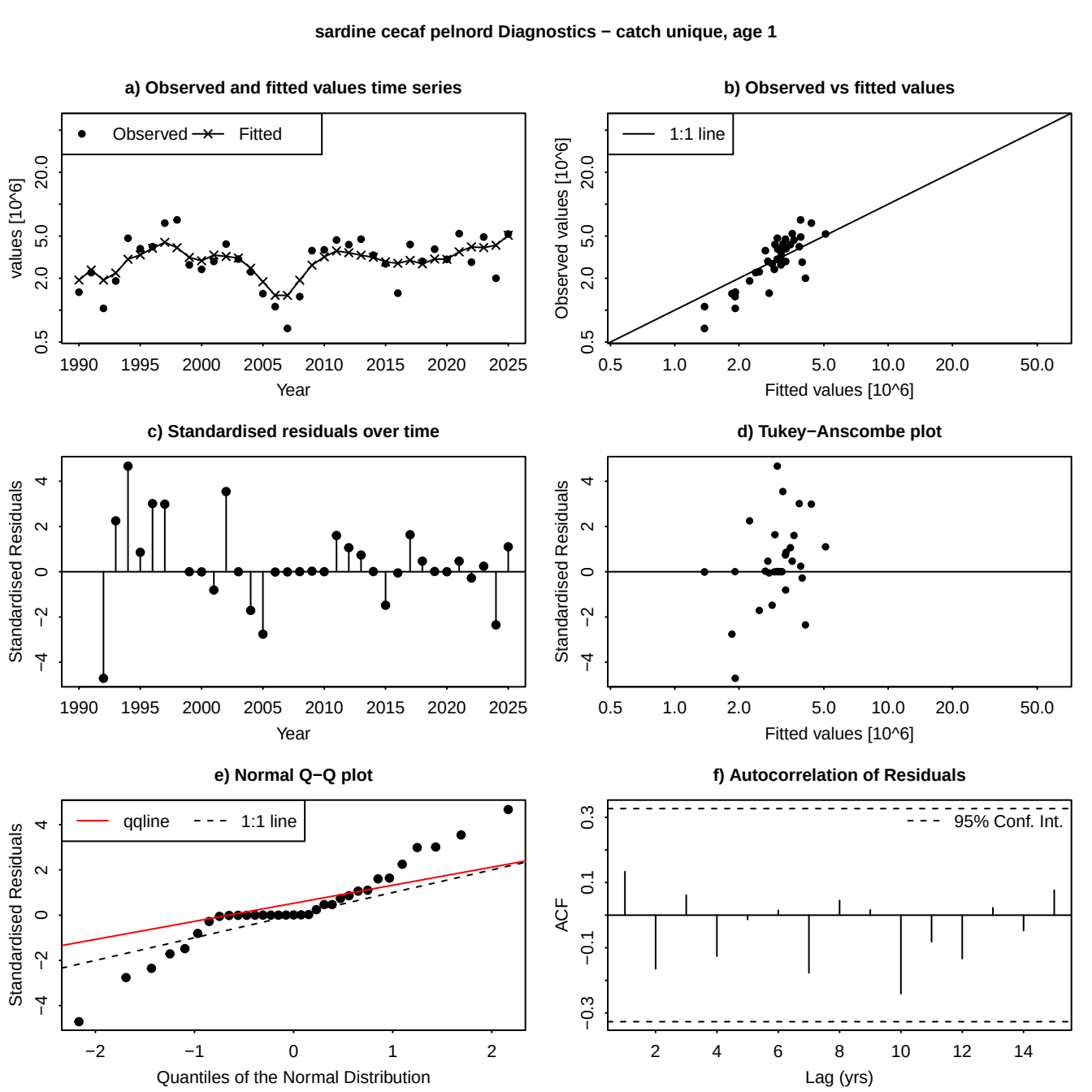
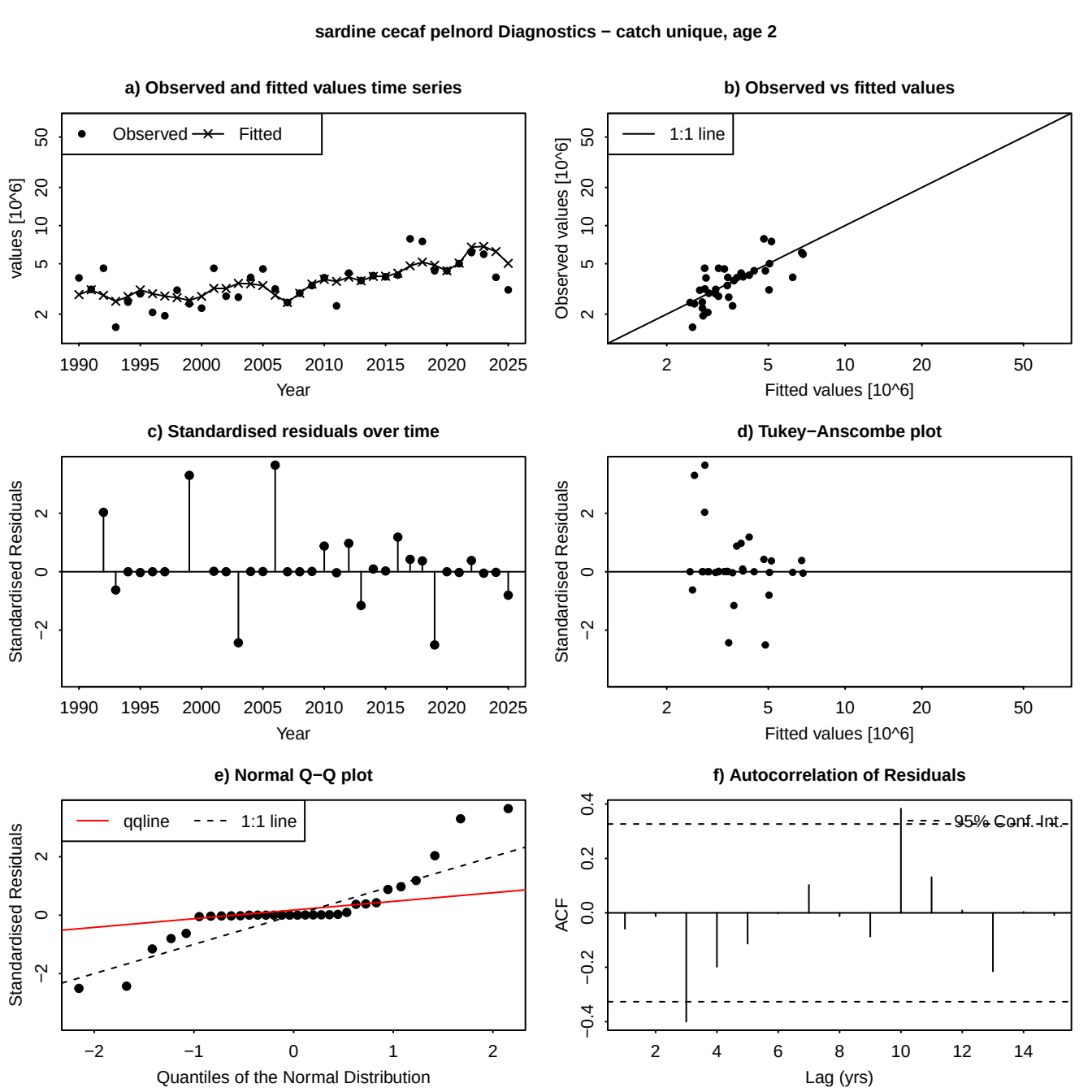
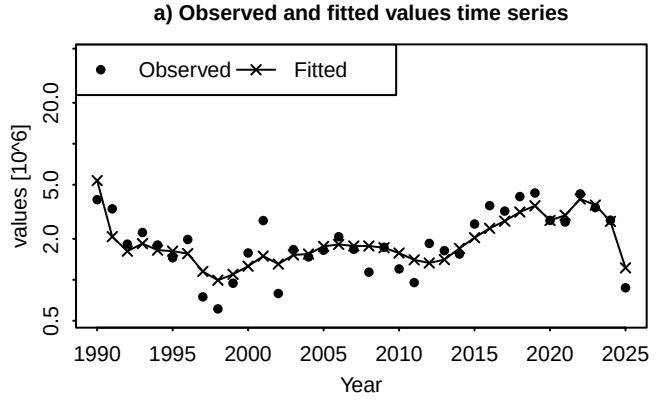
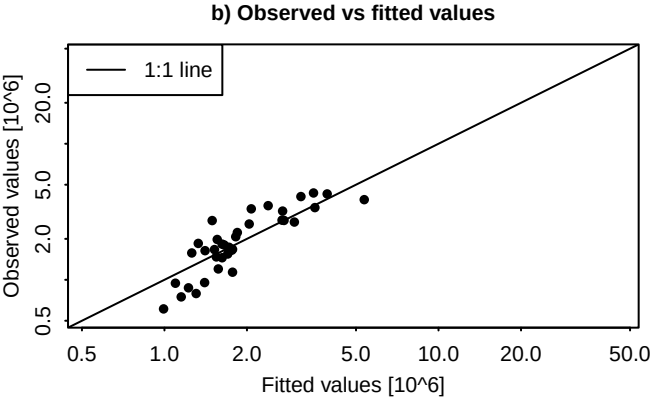

sardine cecaf pelnord Diagnostics − catch unique, age 0
a) Observed and fitted values time series
Observed
Fitted
values [10^6]
0.2
1.0
5.0
20.0
1990
1995
2000
2005
2010
2015
2020
2025
Year
b) Observed vs fitted values
1:1 line
Observed values [10^6]
Fitted values [10^6]
0.2
0.5
1.0
2.0
5.0
10.0
20.0
50.0
c) Standardised residuals over time
Standardised Residuals
−4
−2
0
2
4
Year
d) Tukey−Anscombe plot
e) Normal Q−Q plot
qqline
1:1 line
Quantiles of the Normal Distribution
−2
−1
0
1
2
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.3
−0.1
0.1
0.3
2
4
6
8
10
12
14
Lag (yrs)

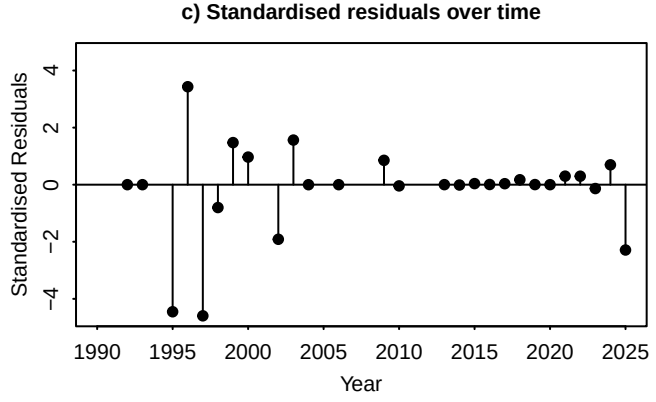

sardine cecaf pelnord Diagnostics − catch unique, age 1
a) Observed and fitted values time series
Observed
Fitted
values [10^6]
0.5
2.0
5.0
20.0
1990
1995
2000
2005
2010
2015
2020
2025
Year
b) Observed vs fitted values
1:1 line
Observed values [10^6]
Fitted values [10^6]
0.5
1.0
2.0
5.0
10.0
20.0
50.0
c) Standardised residuals over time
Standardised Residuals
−4
−2
0
2
4
Year
d) Tukey−Anscombe plot
e) Normal Q−Q plot
qqline
1:1 line
Quantiles of the Normal Distribution
−2
−1
0
1
2
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.3
−0.1
0.1
0.3
Lag (yrs)
2
4
6
8
10
12
14

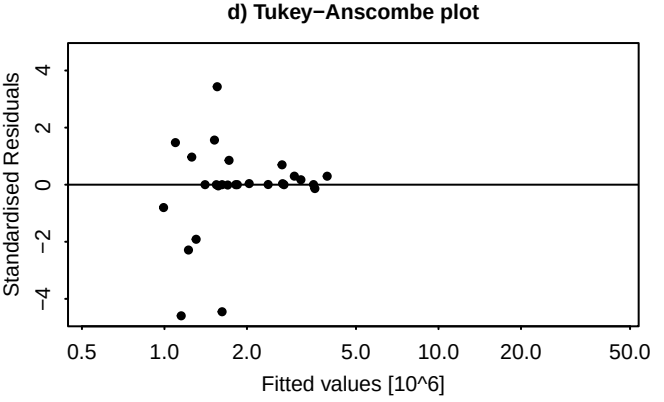

sardine cecaf pelnord Diagnostics − catch unique, age 2
a) Observed and fitted values time series
Observed
Fitted
values [10^6]
Year
b) Observed vs fitted values
1:1 line
Observed values [10^6]
Fitted values [10^6]
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values [10^6]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

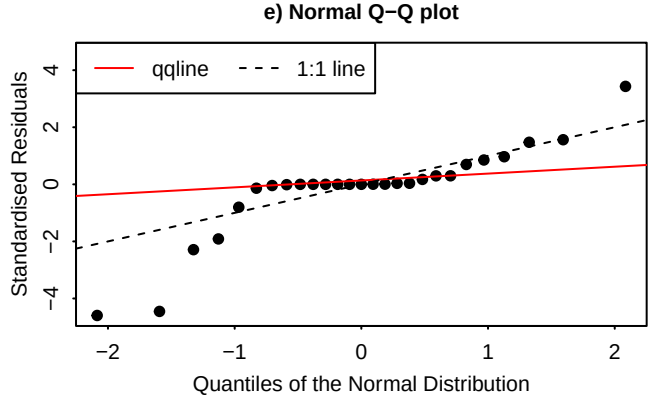
sardine cecaf pelnord Diagnostics − catch unique, age 3

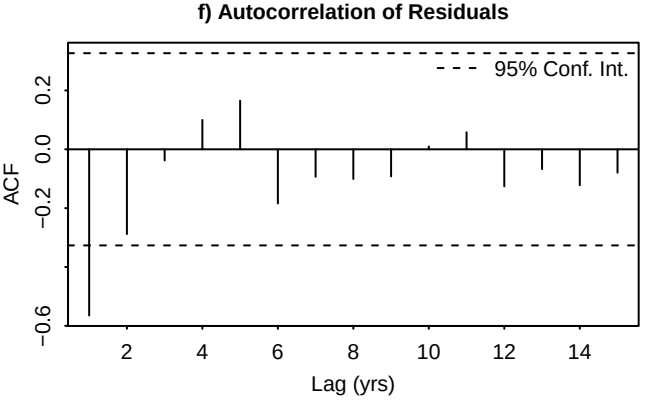


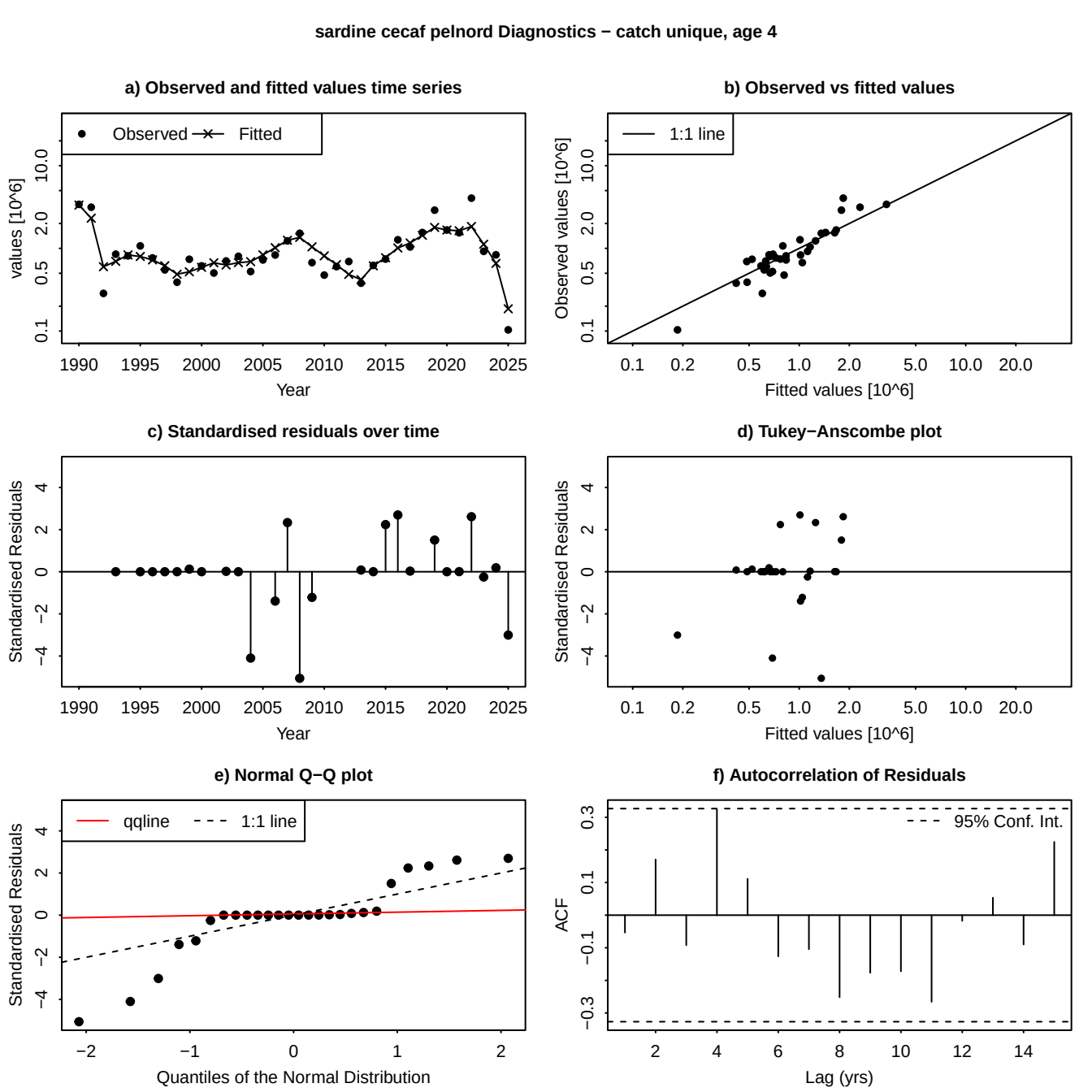




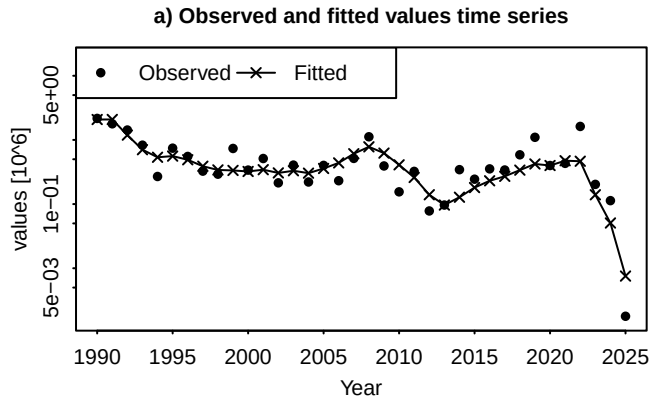


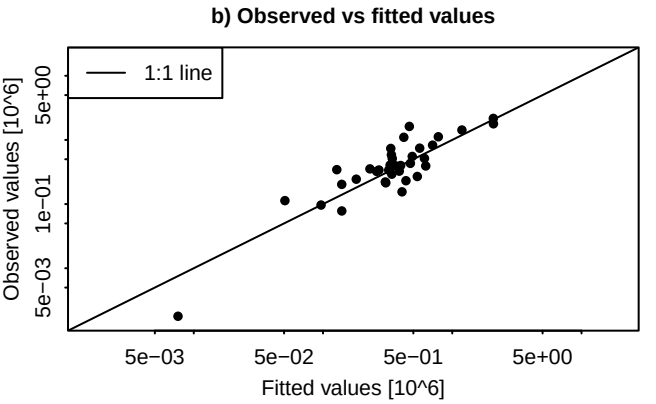


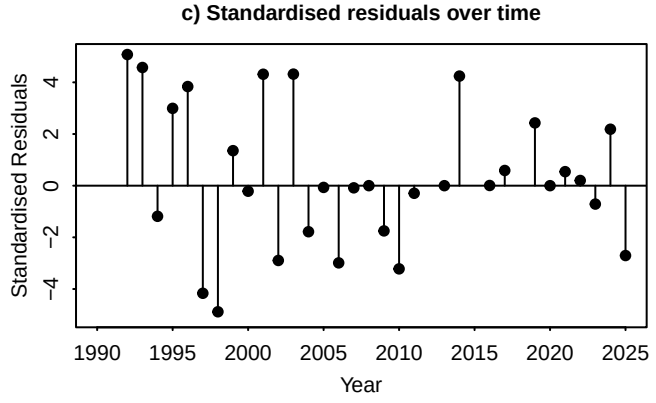


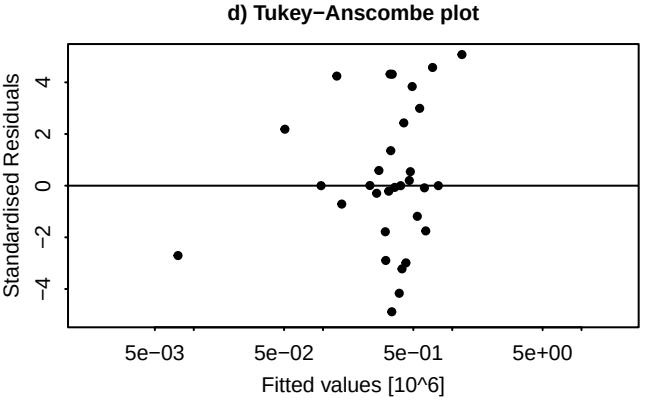

sardine cecaf pelnord Diagnostics − catch unique, age 4
a) Observed and fitted values time series
Observed
Fitted
values [10^6]
0.1
0.5
2.0
10.0
1990
1995
2000
2005
2010
2015
2020
2025
Year
b) Observed vs fitted values
1:1 line
Observed values [10^6]
Fitted values [10^6]
0.1
0.2
0.5
1.0
2.0
5.0
10.0
20.0
c) Standardised residuals over time
Standardised Residuals
−4
−2
0
2
4
Year
d) Tukey−Anscombe plot
e) Normal Q−Q plot
qqline
1:1 line
Quantiles of the Normal Distribution
−2
−1
0
1
2
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.3
−0.1
0.1
0.3
2
4
6
8
10
12
14
Lag (yrs)

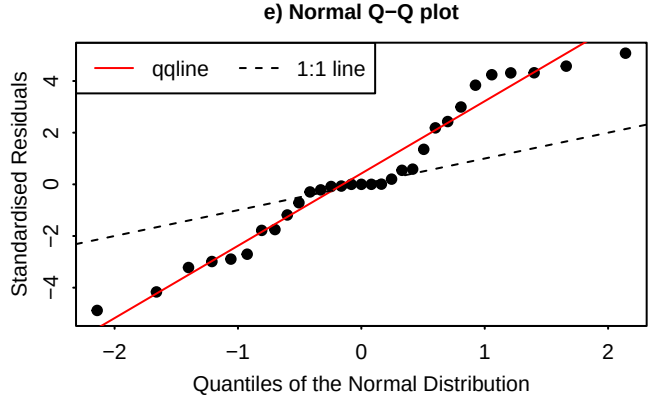
sardine cecaf pelnord Diagnostics − catch unique, age 5

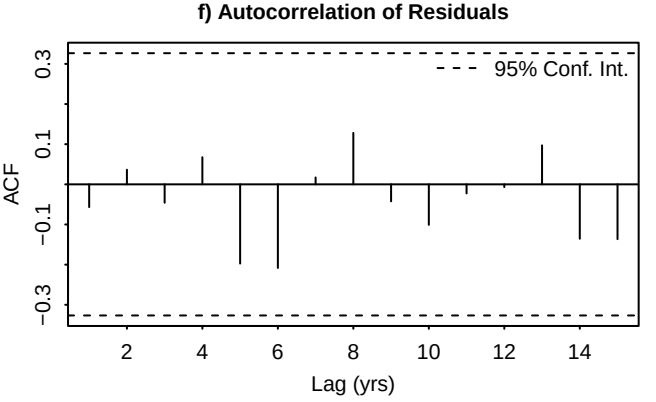


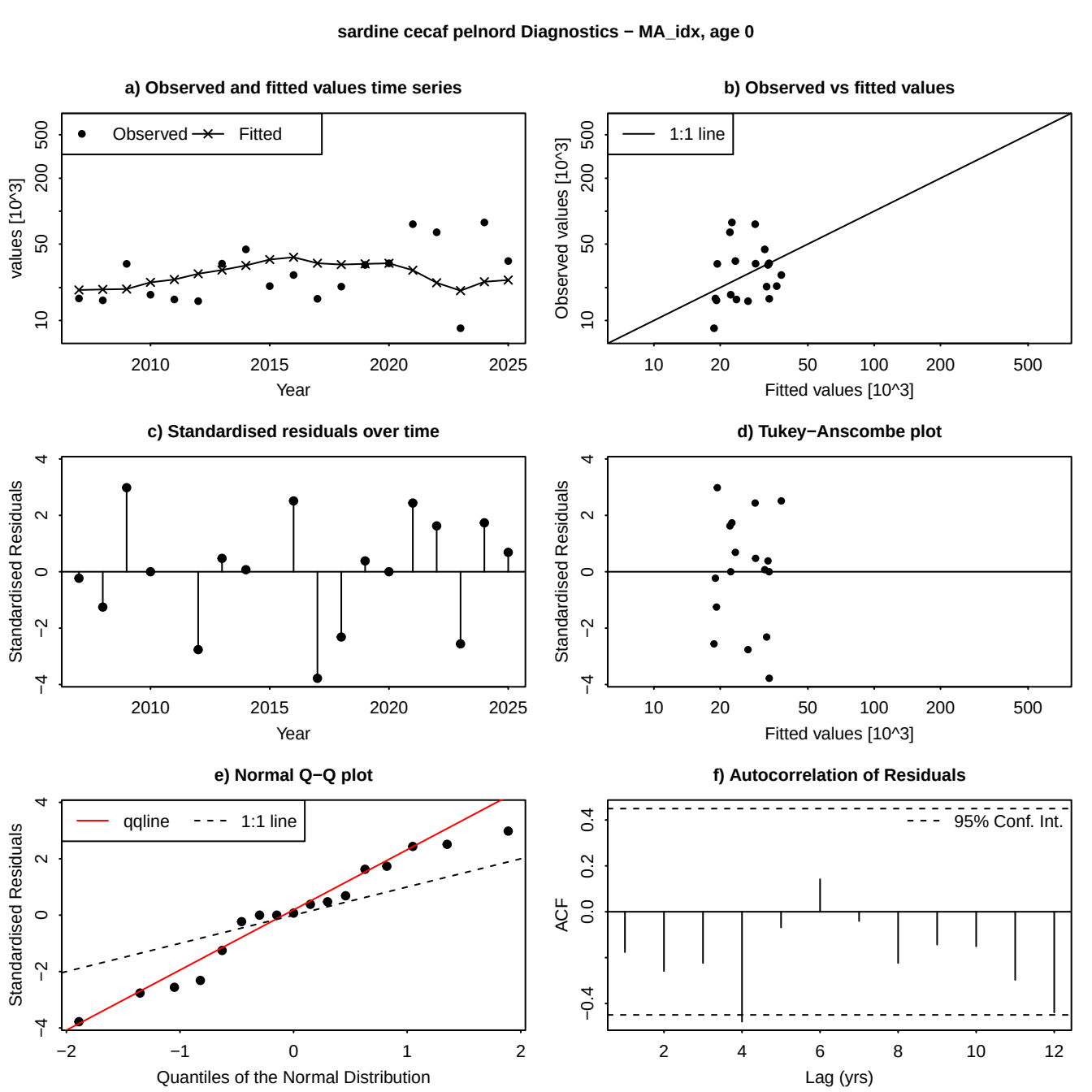


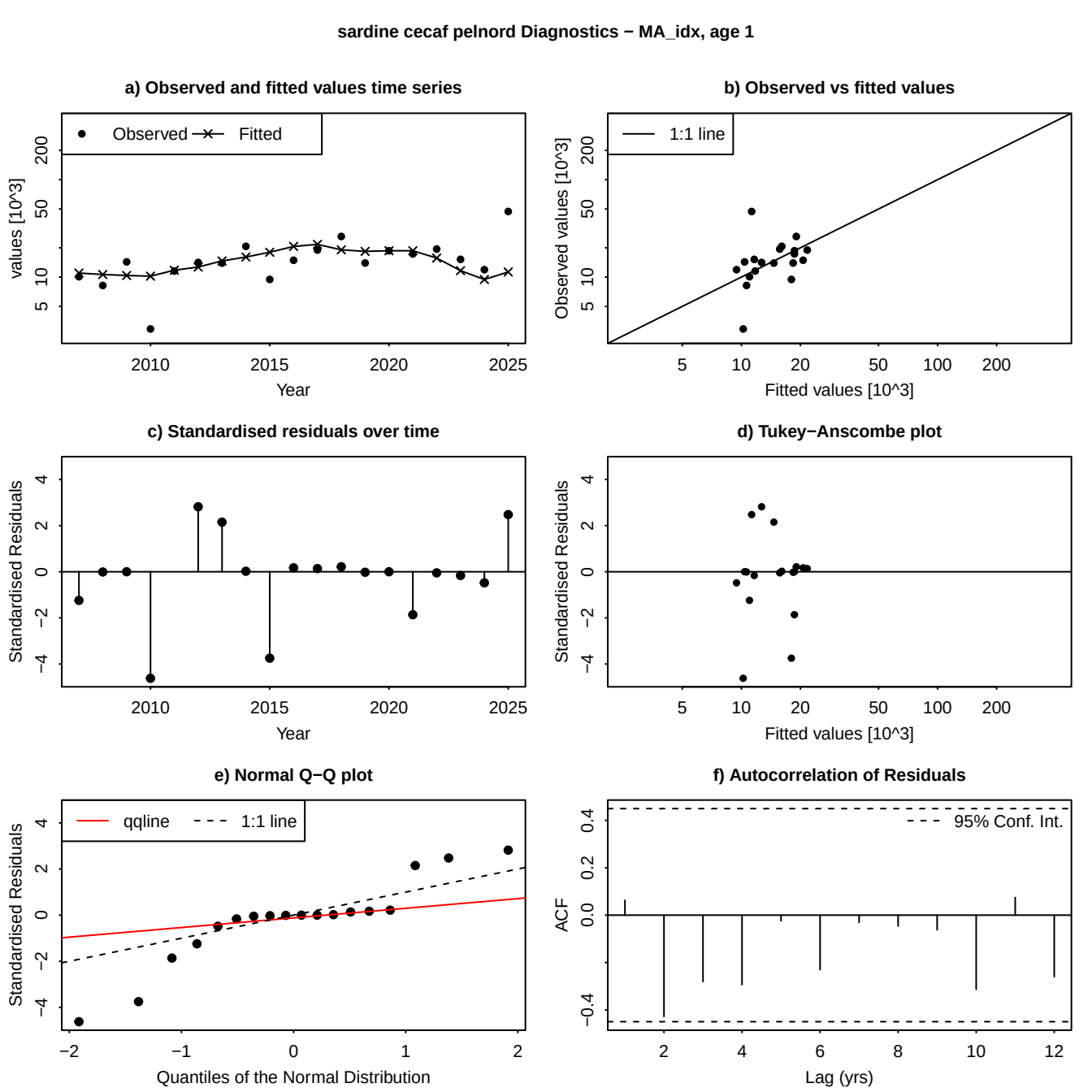


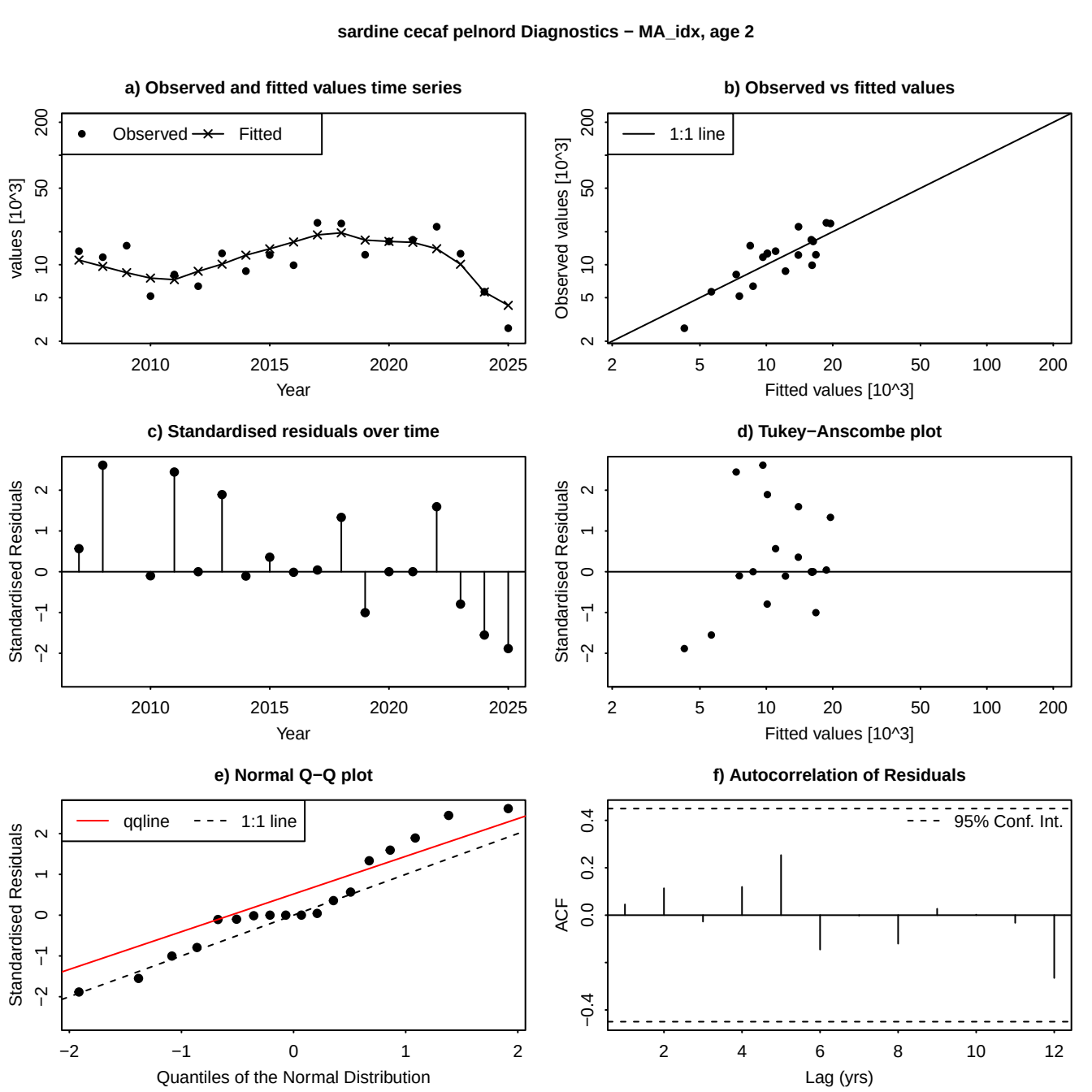


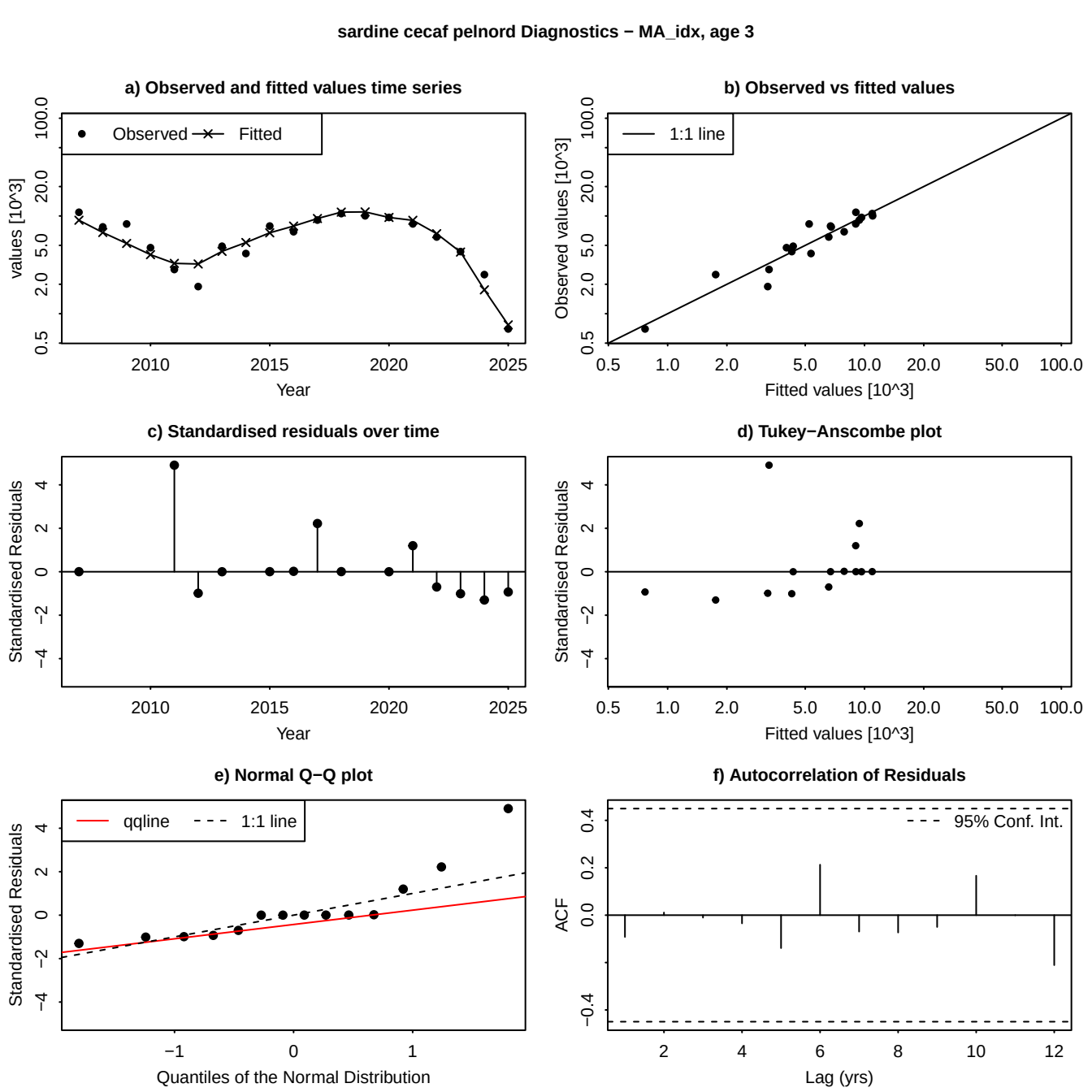


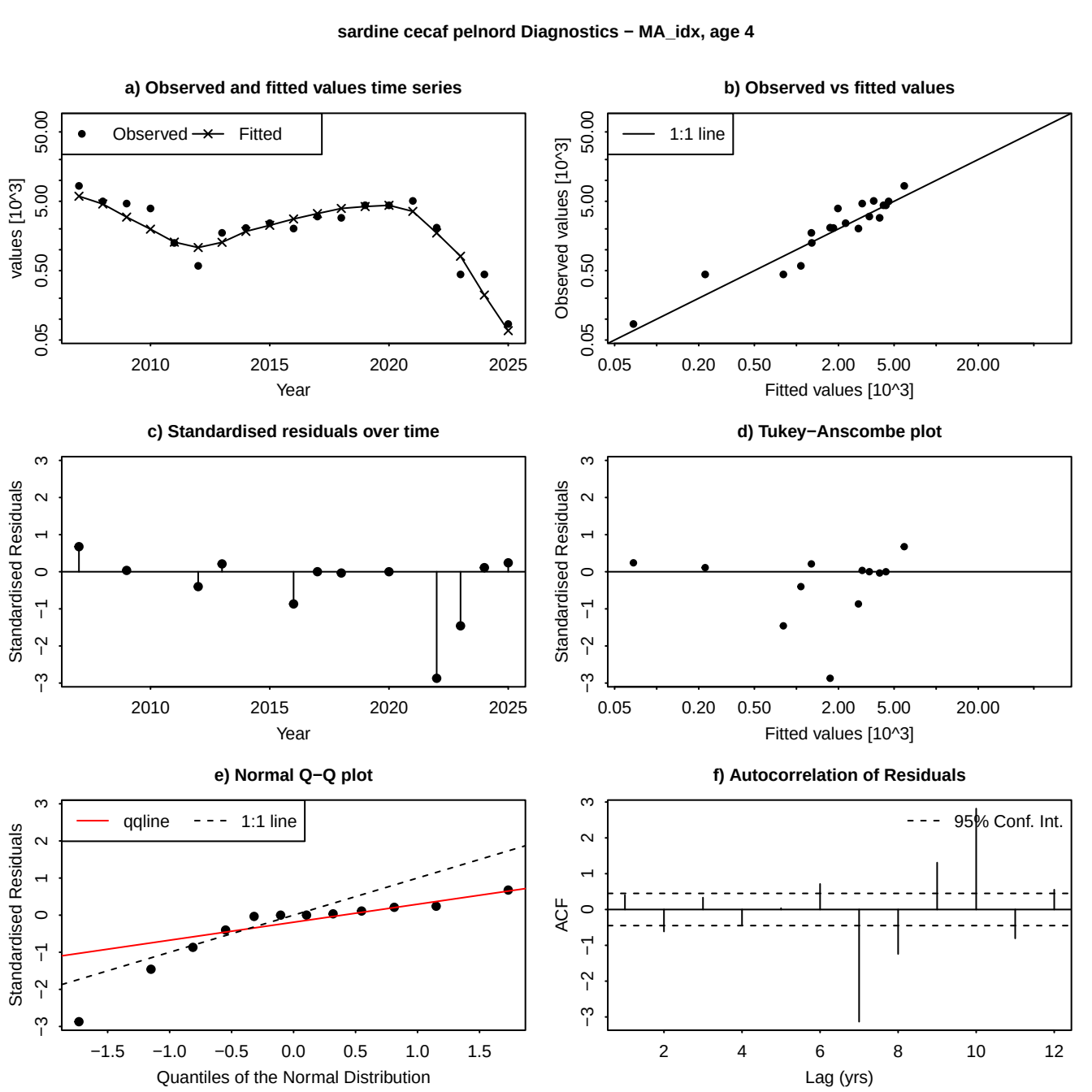


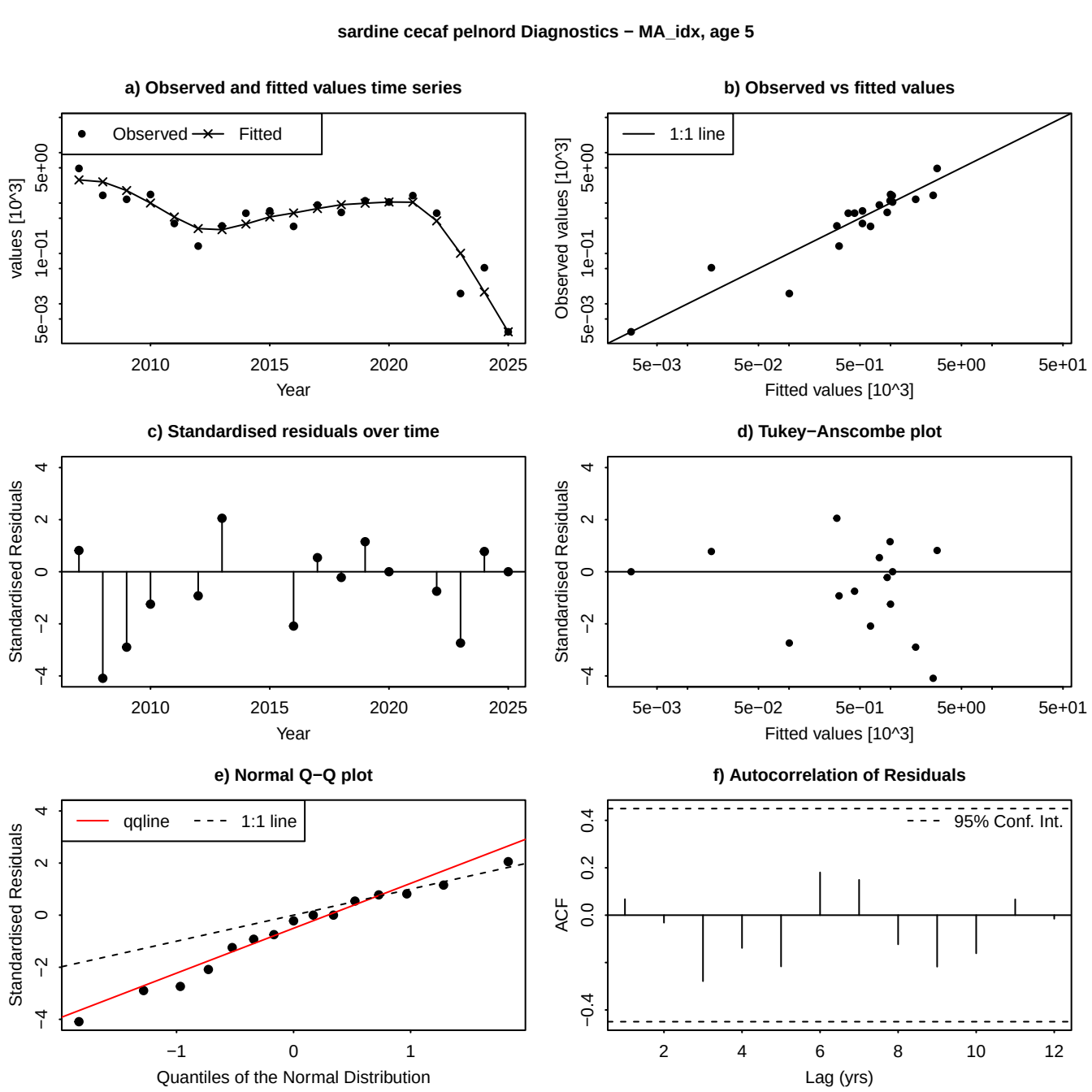

sardine cecaf pelnord Diagnostics − MA_idx, age 0
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
10
50
200
500
2010
2015
2020
2025
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
Fitted values [10^3]
10
20
50
100
200
500
c) Standardised residuals over time
Standardised Residuals
−4
−2
0
2
4
Year
d) Tukey−Anscombe plot
e) Normal Q−Q plot
qqline
1:1 line
Quantiles of the Normal Distribution
−2
−1
0
1
2
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.4
0.0
0.2
0.4
Lag (yrs)
2
4
6
8
10
12


sardine cecaf pelnord Diagnostics − MA_idx, age 1
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
5
10
50
200
2010
2015
2020
2025
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
Fitted values [10^3]
5
10
20
50
100
200
c) Standardised residuals over time
Standardised Residuals
−4
−2
0
2
4
Year
d) Tukey−Anscombe plot
e) Normal Q−Q plot
qqline
1:1 line
Quantiles of the Normal Distribution
−2
−1
0
1
2
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.4
0.0
0.2
0.4
Lag (yrs)
2
4
6
8
10
12

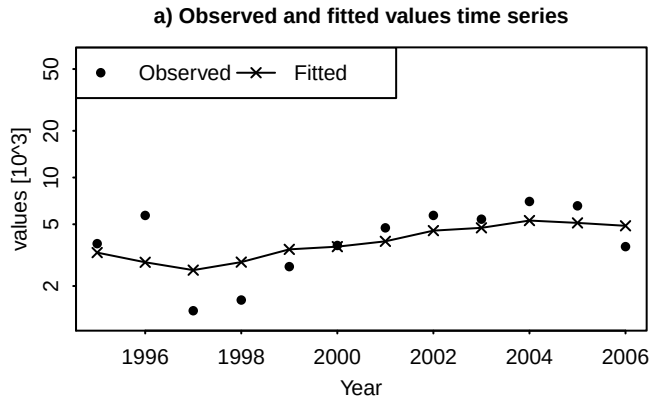

sardine cecaf pelnord Diagnostics − MA_idx, age 2
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

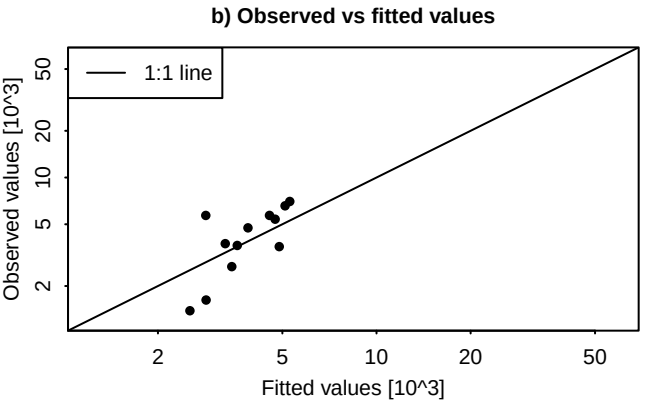

sardine cecaf pelnord Diagnostics − MA_idx, age 3
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

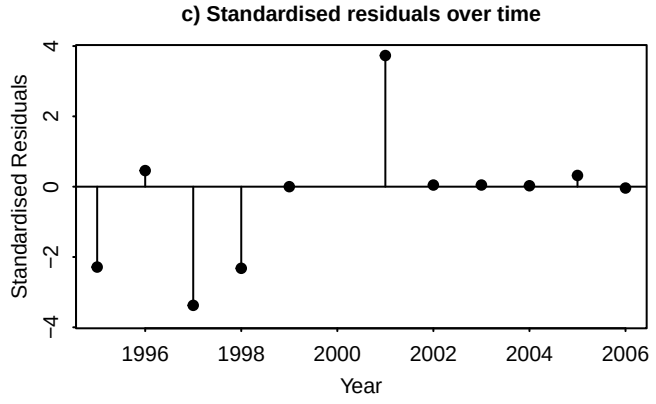

sardine cecaf pelnord Diagnostics − MA_idx, age 4
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
50.00
5.00
0.50
0.05
2010
2015
2020
2025
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
Fitted values [10^3]
0.05
0.20
0.50
2.00
5.00
20.00
c) Standardised residuals over time
Standardised Residuals
3
2
1
0
−1
−2
−3
d) Tukey−Anscombe plot
e) Normal Q−Q plot
qqline
1:1 line
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
2
4
6
8
10
12
Lag (yrs)

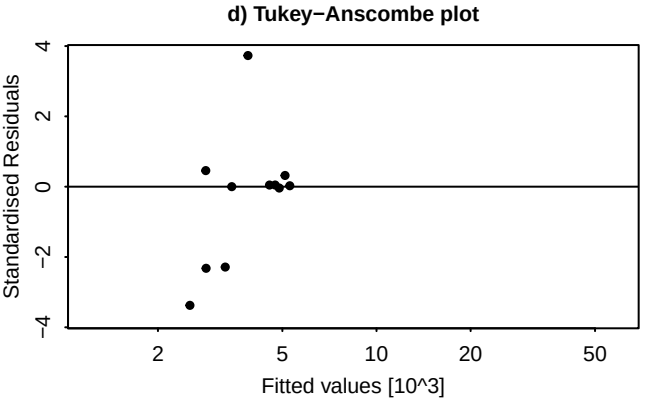

sardine cecaf pelnord Diagnostics − MA_idx, age 5
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
5e−03
1e−01
5e+00
2010
2015
2020
2025
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
Fitted values [10^3]
5e−03
5e−02
5e−01
5e+00
5e+01
c) Standardised residuals over time
Standardised Residuals
−4
−2
0
2
4
Year
d) Tukey−Anscombe plot
e) Normal Q−Q plot
qqline
1:1 line
Quantiles of the Normal Distribution
−1
0
1
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.4
0.0
0.2
0.4
Lag (yrs)
2
4
6
8
10
12

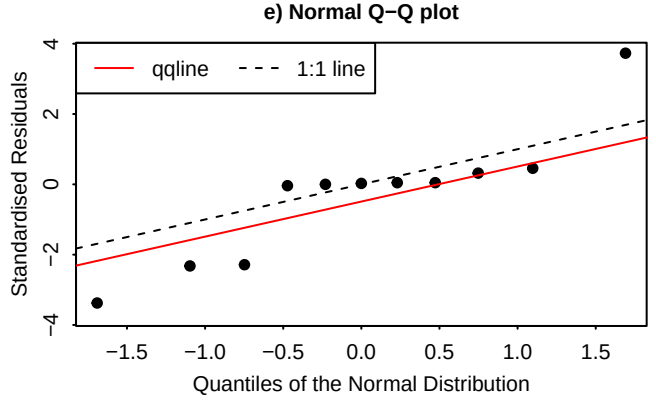
sardine cecaf pelnord Diagnostics − MA_ssb, age −1

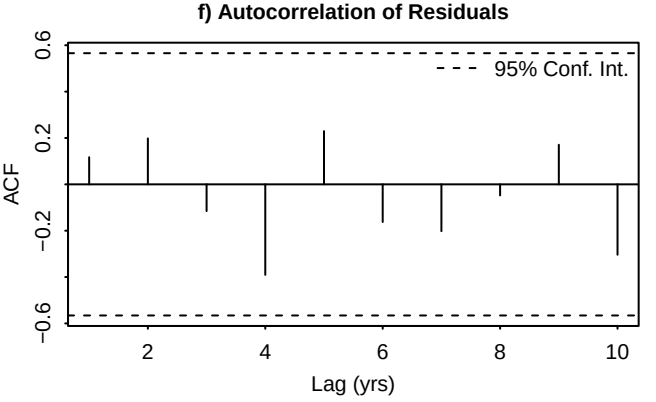


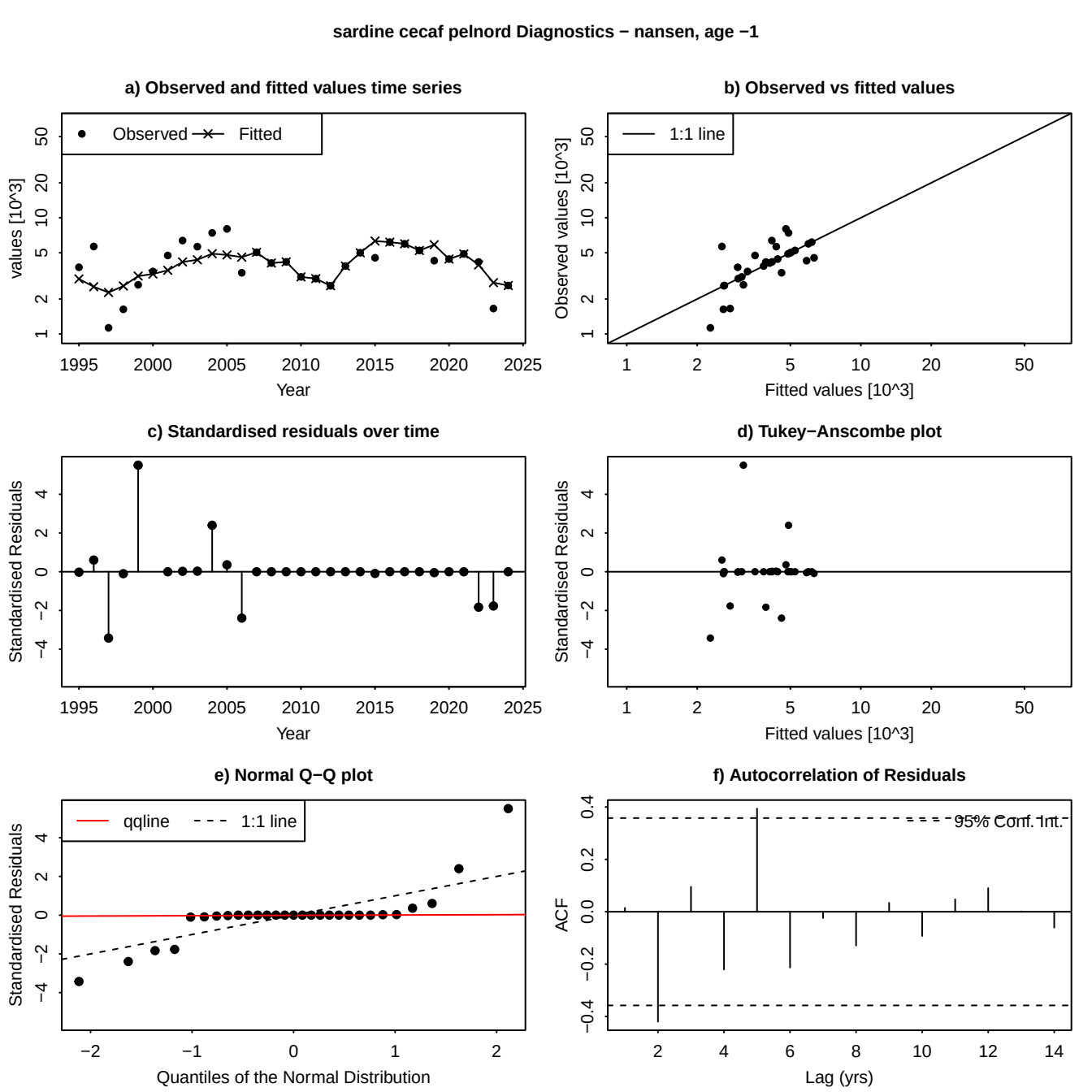











sardine cecaf pelnord Diagnostics − nansen, age −1
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)